# Cap and Trade with Imperfect Hedging

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# Cap and Trade

Emissions externalities + no other friction ⇒ Cap and trade with emissions caps reflecting the SCC achieves first best

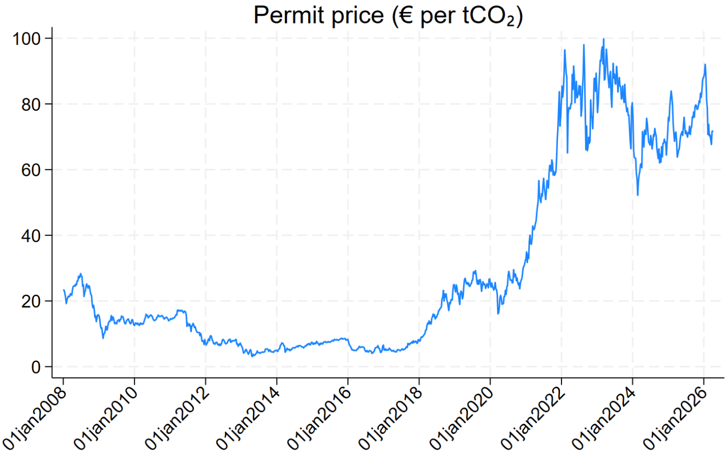
23% of global emissions covered by cap and trade in 2025

SCC is highly uncertain due uncertainty about model structure, parameters, and preferences

> *"The range of estimates is large (...) In the past 10 years, estimates of the Social Cost of Carbon have increased from \$9 per ton CO2 to \$40 for a high discount rate and from \$122 per ton CO2 to \$525 for a low discount rate."* (Tol, 2023)

⇒ Optimal cap is uncertain ⇒ Emitters face transition risk

# Example: EU ETS



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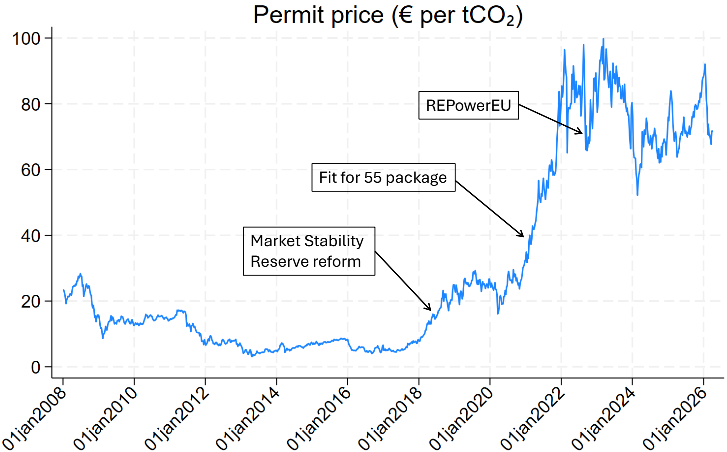

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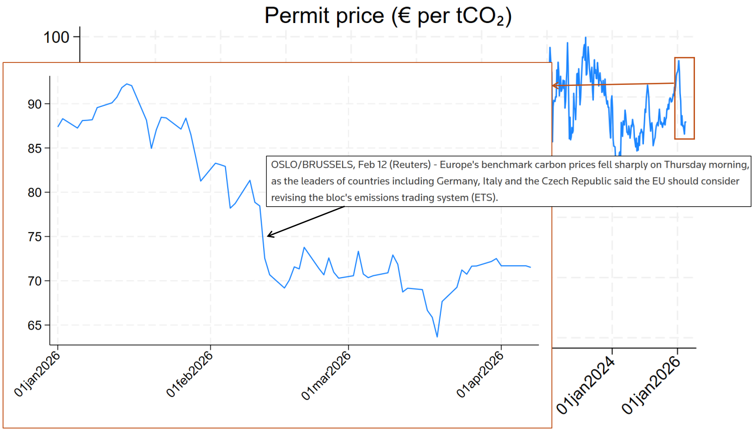


Fuchs, Stroebel & Terstegge (2026): Options-implied volatility on permit price increases around major climate policy events

# Example: EU ETS

Annualized volatility of compliance cost $\simeq$ 40% of EU emitters' pre-tax profits

# Hedging Under Financial Constraints

Price risk creates demand for hedging

Draghi Report (2024):

> *"The hedging component (i.e., removing carbon price uncertainty) could also be fulfilled by sufficient ex ante purchases of ETS allowances as the later are 'bankable' (i.e., unused allowances can be saved for later use). Frontloading purchases of ETS allowances would, however, require up-front financing and may hit the financing constraints of companies."*

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**This Paper:** How do financial frictions affect (1) hedging/pricing of permits and (2) the optimal policy?

- Evidence of hedging/pricing distortions from the EU ETS
- Model with financial friction rationalizing these facts
- Determine second best and its implementation

# Road Map

1. Stylized Facts from the EU ETS

2. Model

# EU ETS

Started in 2008 (after experimental phase 2005-2007)

Firms must surrender permits to cover their yearly GHG emissions

$\simeq$ 50% of permits auctioned, 50% freely allocated

Secondary trading: spot (one-day ahead futures), futures (most liquid is next December), and options

Market participants: emitters and financial firms

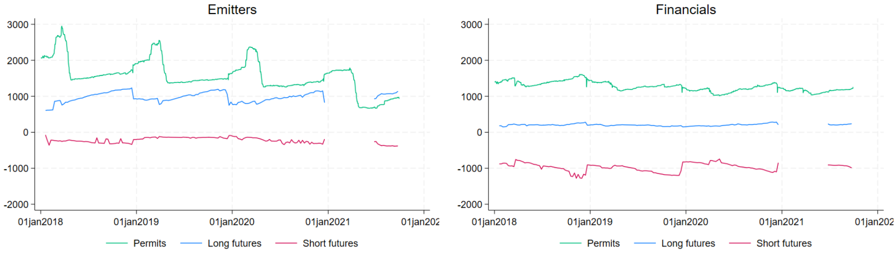
# Fact 1: Risk

- Permit price volatility is high and represents 40% of emitters' pre-tax profits

# Fact 2: Hedging

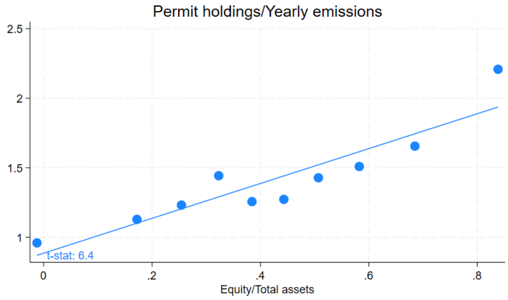
- Emitters buy spot and futures
- Financials sell futures and hedge by buying spot

Holdings of permits and futures (million $tCO_2$)



# Facts 3: Financial Constraints and Hedging

a. Less capitalized emitters hold less permits




Buying permits requires upfront financing $\Rightarrow$ costly for financially constrained firms

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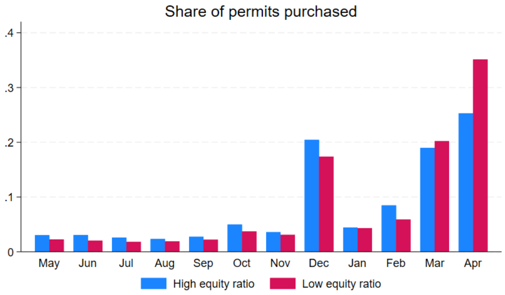
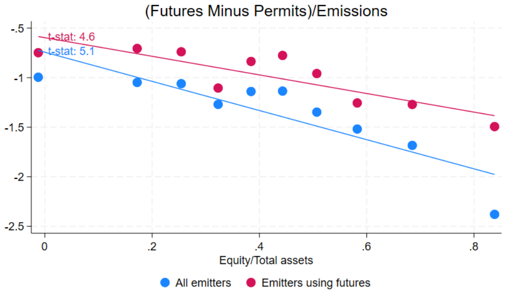


# Facts 3: Financial Constraints and Hedging

a. Less capitalized emitters hold less permits

b. Less capitalized emitters delay permit purchases

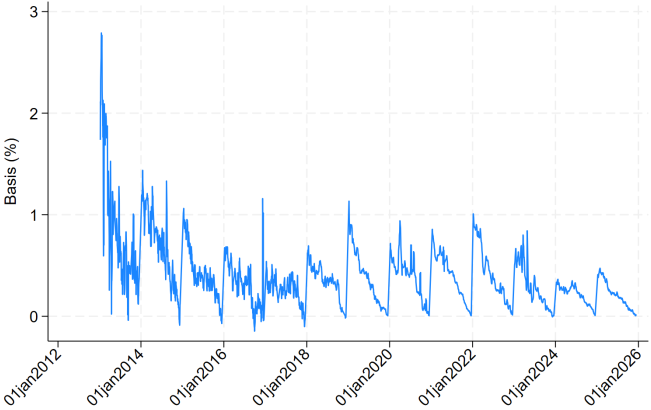
c. Less capitalized emitters buy futures rather than spot




Buying spot requires upfront financing

# Fact 4: Basis

Permit futures trade at a discount relative to spot




Financial constraints $\Rightarrow$ permits costlier than futures

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# Model

Two dates $t = 0, 1$. State realized at $t = 1$: high cap ($s = h$) or low cap ($s = \ell$)

Two types of agents: Emitters and Financials

At $t = 0$:

- Government issues $\overline{n}_0$ permits and rebates proceeds to emitters
- Emitters and financials trade permits and Arrow securities for each state
- Emitters' budget constraint: $$\underbrace{p_0 n_E}_{\text{permits holdings}} + \underbrace{\sum_{s=\ell,h} q_s a_{Es}}_{\text{Arrow sec. holdings}} \leq \underbrace{p_0 \overline{n}_0}_{\text{rebate}}$$

# Model

At $t = 1$ in state $s \in \{h, \ell\}$:

Government issues $\overline{n}_s$ permits, $\overline{n}_\ell < \overline{n}_h$

Emitters choose emissions $e_s$ to produce $f(e_s)$, $f' > 0$, $f'' < 0$

Financials endowed with $y$

E and F consume, utility $u(c_{is}) - \delta_s \overline{e}_s$, social cost of aggregate emissions $\delta_s \overline{e}_s$

Emitters' budget: $\underbrace{c_{Es}}_{\text{cons.}} \leq \underbrace{f(e_s)}_{\text{prod.}} - \underbrace{p_s e_s}_{\text{cost of permits}} + \underbrace{p_s \overline{n}_s}_{\text{rebate}} + \underbrace{p_s n_E}_{\text{permits held}} + \underbrace{a_{Es}}_{\text{Arrow held}}$

Despite full rebate, emitters face risk of low production (due to low cap)

$\Rightarrow$ In equilibrium emitters purchase insurance against low cap from financials (Fact 2)

# Incentive Constraint

- Agents can abscond with fraction $\theta$ of their time 1 resources

$$c_{Es} \geq \theta \left[ f(e_s) + p_s n_E \right] \qquad c_{Fs} \geq \theta \left[ y + p_s n_F \right]$$

$\Rightarrow$ Limits how much insurance can be provided

- Low cap: emitters produce little, financials transfer resources to emitters but not as much as in the first best when IC binds

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$\Rightarrow$ Extension: heterogeneous emitters with different endowment at time 0

- Emitters with lower endowment $\Rightarrow$ higher liabilities $\Rightarrow$ tighter IC $\Rightarrow$ buy less permits at time 0 (buy permits later) (Facts 3)

# Optimal Policy

- Planner chooses emissions and consumption as a function of the SSC
- Maximizes
$$\sum_{i=E,F} \alpha_i E\left[u(c_{is}) - \delta_s e_s\right]$$

  subject to resource constraint $c_{Es} + c_{Fs} \leq f(e_s) + y$

  and incentive constraints $\begin{array}{l} c_{Es} \geq \theta\, f(e_s) \\ c_{Fs} \geq \theta\, y \end{array}$

- Uncertain SCC: $\delta_\ell > \delta_h$
- Compare the constrained optimum (with IC) to the first best (without IC)

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Result 1: Constrained-optimal emissions are **less responsive** to the SCC than in the first best

- Binding ICs limit risk sharing $\Rightarrow$ planner reduces transition risk

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Result 2: Constrained-optimal emissions depend on agents' **Pareto weights** (e.g. political weights), whereas first-best emissions don't

- First best: emissions only pinned down by SCC (while Pareto weights pin down transfers)
- Constrained optimum: transfers are constrained $\Rightarrow$ transfer utility by distorting emissions

# Implementation

Result 3: Constrained optimum can be implemented with **cap and trade** and transfers

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Result 4: Requires zero permits issuance at $t = 0$

- Because storage of permits from $t = 0$ to $t = 1$ tightens IC
- **No front-loading** of permits issuance is consistent with EU's policy

# Wrap-Up

- Cap and trade $\Rightarrow$ transition risk due to uncertainty in SCC $\Rightarrow$ demand for hedging

- Financial constraints $\Rightarrow$ limited hedging $\Rightarrow$ distorts optimal policy
  - Cap-and-trade remains optimal with zero front-loading of permits issuance
  - Optimal cap is less responsive to SCC
  - Optimal cap depends on political weights